

Chapter Title: What Do We Mean by Poverty?

Book Title: Measuring Poverty around the World

Book Author(s): Anthony B. Atkinson

Book Editor(s): John Micklewright, Andrea Brandolini

Published by: Princeton University Press. (2019)

Stable URL: <https://www.jstor.org/stable/j.ctvc77fd6.8>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



Princeton University Press is collaborating with JSTOR to digitize, preserve and extend access to *Measuring Poverty around the World*

What Do We Mean by Poverty?

The “we” in the title of this chapter are the many people who have diverse concepts of poverty and apply the term in a variety of contexts. Poverty is interpreted in different ways in different countries and in different cultures. Poverty has been approached differently at different times in the past. Poverty is a contested concept, and readers of this book may hold a diversity of views about what they understand it to mean. The different concepts of poverty may each be legitimate and—what concerns me here—may have potentially different implications for how it is measured. There may not therefore be a single answer to the question, what is the extent of poverty? In this chapter, I describe a range of possible conceptual approaches: “political” definitions, subjective assessments, measures grounded in basic needs for consumption, measures derived from the capability approach, and measures based on minimum rights.

Which of these approaches should be adopted is a matter of personal judgement. I am seeking only to offer a menu of choice; there is no “chef’s recommendation.” But I should stress at the outset the importance of distinguishing between the judgements on which people base their own behaviour and the judgements that they would like to see applied in making decisions as a society as a whole. These may well be different. The former may give primacy to the standard of living of their own family, but the latter may lead them to support their local government implementing programmes of assistance to the homeless. There is nothing illogical in such a separation of judgements. Indeed, the separation is an essential feature of theories of justice, such as that of the Harvard philosopher John Rawls in his *Theory of Justice* of 1971. Rawls proposes that principles of justice should be established by people considering, not their actual circumstances, but how they would make judgements behind a “veil of ignorance,” not knowing where they personally would end up. While I do not resort to such a rhetorical device, the discussion of different concepts of poverty is concerned with the way in which we would like to see our society—local, national, or global—approach the issue.

POLITICAL DEFINITIONS OF POVERTY AND SUBJECTIVE PERCEPTIONS

The distinction between the first concept considered—a “political definition”—and the remainder of the chapter is neatly illustrated by the contrast between

the *Merriam-Webster Dictionary* definition of the “poverty line” as “a level of personal or family income below which one is classified as poor according to governmental standards” with that in the *Oxford English Dictionary*, according to which the poverty line is “the estimated minimum level of income needed to secure the necessities of life.” In this section of the chapter, I begin by exploring the implications of defining poverty in terms of the standard adopted explicitly or implicitly by political authorities, whether at the local, regional, national, or global level.

Political action and assessing performance

Why should we pay attention to the political role of concepts of poverty? The first reason is that, as already observed in chapter 1, statements about the extent of poverty have served a significant function in motivating political action. A story from the early life of Sir Winston Churchill illustrates the political salience of statistical evidence about poverty in Britain. In 1901, Churchill reported that he had been studying Seebohm Rowntree’s account of poverty in York: “I have been reading a book which has fairly made my hair stand on end [which] deals with poverty in the town of York. It is found that the poverty of the people of that city extends to nearly one-fifth of the population. . . . That I call a terrible and shocking thing” (reported in Jenkins 2001, p. 81). This account of British poverty contributed to Churchill leaving the Conservative Party and joining the 1906 Liberal government. As a Liberal minister, he introduced labour exchanges and unemployment insurance. As I shall demonstrate at a number of points, this is just an early example of how statistics have played an essential role in motivating action. The initiation of the War on Poverty in the United States by President Johnson following President Kennedy’s concerns, described in chapter 1, is a second powerful example.

Statistical evidence about the extent and nature of poverty has been a major factor influencing political action in part because the existence of poverty reveals policy failure. The statistics are a performance indicator. The rediscovery of poverty in Europe was a wake-up call because it destroyed the comfortable belief that poverty had been eliminated in rich countries by a combination of economic growth, full employment, and the welfare state. The promise of an income floor, a guaranteed national minimum, was taken by those concerned with poverty as an implicit standard by which to judge performance. If people are living below this level, then this is a failure of the postwar welfare state. Coming to the present time, the minimum living standard guarantee, or *dibao*, programme introduced by China in urban areas in the 1990s and in the 2000s in rural areas provides a reference point when considering the official poverty lines.

The performance indicator role has been resisted by some statistical agencies. In Canada, it has been argued that “the low-income statistics are not

intended to provide an indication of the success or failure of specific programs designed to assist the poor” (Murphy, Zhang, and Dionne 2012, p. 6n). The former chief statistician of Canada, Ivan Fellegi, has stated categorically that “the essentially political nature of such estimates render it inappropriate for a Statistical Agency to make such judgements” (Fellegi 1997). It is, however, not clear that other statistics produced by his agency are free from political judgements. Would anyone replace “low income statistics” with “national accounts” and claim that “the national accounts statistics are not intended to provide an indication of the success or failure of specific programs designed to raise the growth rate”? Yet anyone with experience of the construction of national accounts knows that apparently technical choices can have major political implications. When I reviewed the measurement of government output in the UK national accounts, the chief economist on my team was summoned to a meeting with members of the Cabinet to discuss the implications for the measured productivity of the government sector. I refused to attend because I wished to form my own independent view, but then, I had the luxury of being an academic, not an official statistician.

The role of performance monitoring has acquired much greater salience with the adoption, nationally and globally, of targets for the reduction or elimination of poverty. Statisticians are asked to report on the progress made towards the Millennium Development Goals and now the Sustainable Development Goals. Reports at the global level, such as the *Global Monitoring Report* prepared by the International Monetary Fund and the World Bank, assemble key data on progress towards the agreed targets. The foreword to the 2015–2016 report stated that “this is a pivotal year for global development. The Millennium Development Goals (MDGs) have guided countries and partners over the last 15 years in improving the living conditions of the poor. We are now transitioning to the Sustainable Development Goals (SDGs), a new set of global targets that embrace economic, social, and environmental priorities through 2030” (World Bank Group 2016, p. ix).

The findings are watched nervously at the national level. The 2015–2016 report, for instance, picked out Niger as the country with the highest rate of multidimensional poverty and identified Pakistan as a country where multidimensional poverty, at 44 per cent in 2013–2014, was much higher than the income-based poverty rate, which in 2010 was 13 per cent. At a regional level, the measurement of performance has in the past led to stark conclusions. The Australian Agency for International Development (AusAID) produced in 2009 a report “intended to help governments design policies and programs that hasten progress towards the MDGs,” but its assessment was sombre: “The report finds that the Pacific as a whole is significantly off track to meet the MDGs by 2015. Some countries are translating economic growth into reducing poverty and meeting the MDGs. However, overall the Pacific is stalling and falling short of the goals” (AusAID 2009, p. 1). At a national level, the collection of household survey data is seen as a key to monitoring performance. The foreword by

the vice chair of the National Planning Commission in Nepal to a report by the statistical office on poverty trends stated that the “wealth of information contained in this poverty report will greatly support in the monitoring and evaluation of the Tenth Plan/Poverty Reduction Strategy, MDGs and other development projects” (Nepal Central Bureau of Statistics 2005, p. i).

Monitoring is not, however, the preserve of governments: it should also be at the forefront of the activities of those who wish to hold governments to account. It can aid concerned individuals, and it can provide evidence to support the case being made by action groups. At a time when there were no statistics about poverty, critics of the government could refer to personal experience or to case studies. These are often telling, but they are much more persuasive if accompanied with statistical evidence about the extent and depth of the problem. As a member of the Child Poverty Action Group (CPAG) in the United Kingdom, I was involved in producing statistically based evidence that called into question the record of the 1964–1970 Labour government and called for more radical measures to tackle poverty. We were bitterly attacked and later blamed for Labour losing the subsequent general election. I doubt the latter, but the bitterness of the attack owed much to the fact that the evidence could not simply be dismissed.¹

The political nature of poverty measurement is a theme that runs through the book, and it provides the immediate justification for studying poverty as defined by political concerns and agreements. But political influences themselves have more fundamental origins, to which we now turn.

Subjective views about poverty

Political views about the definition of poverty do not arrive simply out of the ether. As the example of Churchill illustrates, politicians are influenced by their experience, and much of that experience comes from talking to people. The views of their citizens—and particularly their voters—can sway them to give prominence to the issue and to consider the nature of poverty. For this reason alone, we should consider the subjective views of those exposed to the risk of poverty. But such views are themselves important in a society that aims at full participation of all its members.

¹*Editors:* This episode is told in a report celebrating the fiftieth anniversary of the CPAG: “It [CPAG] began to campaign much more publicly than before, at the national level, with considerable impact. In May, it issued an election manifesto, *A War on Poverty: Poor Families and the Election*, with press releases headed ‘Poor get poorer under Labour,’ repeating its and other campaigners’ demands. It got good radio, TV and press coverage which impressed politicians. Edward Heath, the Conservative leader, wrote a letter of support to CPAG and repeatedly quoted its claims in campaign speeches attacking Labour. CPAG’s attacks on Labour increased its credibility among Conservatives and more widely, raising its national profile and that of poverty as a political issue, while some blamed it for helping to bring about Labour’s election defeat” (Thane and Davidson 2016, p. 17).

A criticism commonly expressed in submissions to the World Bank's Commission on Global Poverty was that more should be done to solicit the views of the people living in extreme poverty. Such a participatory approach would learn from the experience of the poor in the countries where most of them are to be found. Why, it is asked, is the International Poverty Line designed in an institution based in a country where extreme poverty is assumed to be nonexistent? Why does the discussion seem to be dominated by the research of those who—like the present author—live in high-income countries?

The World Bank has indeed been alive to this criticism. A major feature of the *World Development Report 2000/2001* was the “Voices of the Poor” background study. This was based on a review of participatory studies involving some forty thousand poor people in fifty countries and on a comparative study in 1999 in twenty-three countries engaging about twenty thousand poor people (World Bank 2001, box 1, p. 3). There were a series of books (Narayan, Chambers, et al. 2000; Narayan, Patel et al. 2000; Narayan and Petesch 2002), the first of these recognising that “the development discourse about poverty has been dominated by the perspectives and expertise of those who are not poor” (2000, p. 2). More recent research has demonstrated the potential. Using household survey microdata from Benin (low-income), Mexico (upper-middle-income) and the United Kingdom (high-income), for example, Nandy and Gordon “show how, in each of the countries selected, there is a high degree of consensus about the necessities of life, and that such consensus allows for the identification and establishment of social norms and those individuals/groups unable to meet such norms due to a lack of resources to be identified” (2015, abstract). As put by the Global Coalition to End Child Poverty, it is “only through a better representation and understanding of poverty—including who is poor . . . and how they experience poverty—[that] we can meet the urgent challenges of ending poverty in all its dimensions and reducing inequalities, leaving no one behind” (2015, para. 1.4).

The reference to “all its dimensions” is important. The findings of the “Voices of the Poor” study suggested that poverty was seen as consisting of many interlocking dimensions, where lack of food, poor health and illness, lack of access to public goods, and powerlessness were judged to be more important than monetary poverty. This underlines the significance of introducing non-monetary poverty and of applying multidimensional indicators. As already signalled, these are an essential part of the book. The potential role of the participatory approach in identifying the relevant domains is discussed further in the next chapter.

Subjective views about poverty are undoubtedly a valuable ingredient, and more efforts should be made to develop the approach. For there are limits to what can be learned from “asking people,” and not all views are equally valid. For instance, we may wish to ignore completely expressions of negative feelings towards others, such as the denial of the existence of poverty and hard-

ship. Many years ago, in a more vigorous treatment of welfare economics than typically takes place today, Sir Dennis Robertson dismissed such elements, saying that our judgements should not “be eroded by the gnawings of the green-eyed monster” (1954, p. 678). He was referring to jealousy, but his warning would apply today to views based on religious bigotry and xenophobia.

There are in fact four ways in which subjective views can be applied:

1. Ask people to assess their own poverty status without any reference to a poverty line, and the total in poverty is then those who classify themselves as “living in poverty” (where the exact formulation of the question is clearly crucial).
2. Ask people about the minimum consumption they consider necessary to avoid poverty, and then ask whether or not their own consumption meets that standard.
3. Ask people about the minimum consumption they consider necessary to avoid poverty, and then obtain information about the household’s consumption to determine whether or not their own consumption meets that standard.
4. Ask people where the poverty line, defined in terms of consumption, should be set, where this enters into the determination of the poverty line for the whole population (and no assessment is made of the status of the household interviewed).

These approaches are obviously different. The fourth is closest to the World Bank procedure for counting the poor (used for the estimates of global poverty in figures 1.2 and 1.3 in the last chapter) in that it leads to a population-wide poverty line.² It has indeed been extensively used. In the United Kingdom, the 1983 Breadline Britain study (Mack and Lansley 1985) identified the poor as those suffering from an enforced lack of socially perceived necessities. These necessities were identified by public opinion as possessions and “activities” that every family should be able to afford and that nobody should have to live without. (Only items considered a necessity by at least 50 per cent of the population were retained.)

The other three approaches lead to an assessment at the level of the responding household and no population-wide poverty line. The first makes no explicit reference to consumption but may be closest to the concerns of politicians. The finding that a substantial proportion of the population consider themselves “poor” may gird them for action, regardless of the basis for this subjective valuation. It is striking that in Uganda in the 2012–2013 Household

²*Editors:* The procedure used by the World Bank to arrive at the International Poverty Line of \$1.90 a day is described in chapter 5 and does not draw on subjective views. Tony’s comment here is merely intended to note that this way of applying subjective views does result in a poverty line that can be applied to the whole population, like the World Bank’s.

Survey, 70 per cent of the population classified themselves as “very poor” or “poor,” whereas the rate of monetary poverty according to the national poverty line was 19.7 per cent (Uganda Ministry of Finance 2014, table 2.4). The second approach is like the others in that it is mediated through consumption per head. The third approach differs in that, like the second approach, it is based on a subjective assessment of poverty status, but the subjective assessment is contrasted with *objective* measurement from survey information recorded by external observers. In all of the first three approaches, the poverty line being applied varies from one household to another because of their different subjective views. Any reported poverty line would be an average over respondents. It should equally be noted that both the first and second approaches can be applied to individuals within the household and may provide a source of evidence about within-household inequality, a subject considered further in the next chapter.

In interpreting the information obtained from participatory studies, the differences are important. As is well known (for example, from the literature on the use of reported happiness), the responses may reflect both adaptation and aspiration. Where people have adjusted to low levels of living, they may have adapted to these levels and regard mere survival as escaping poverty. In the opposite direction, responses may be influenced by rising aspirations, where people above the objective poverty line see themselves falling behind rising living standards elsewhere. In both cases, there is a potential bias, in the first case downward and in the second case upward. Moreover, the bias may differentially affect different groups. Remote rural communities, for instance, may be more likely to be affected by adaptation and their extreme poverty more likely to be understated.

In terms of concrete implementation, there are at least two possible roles for the collection of information by participatory input. The first application is to the choice of alternative poverty lines. For many years the World Bank has recognised multiple poverty lines. The 1990 *World Development Report* applied a line of \$275 a year for the “extremely poor” and \$370 a year for the “poor” (World Bank 1990, table 2.1). The 2000–2001 report, in addition to the \$1 a day line (by then, strictly, \$1.08 a day), showed the results of an upper poverty line of \$2 a day, “reflecting poverty lines more commonly used in lower-middle-income countries” (World Bank 2001, p. 17).³ Quite a number of countries make use of multiples of the International Poverty Line (now household consumption per head of PPP\$1.90 a day—see chapter 1). Employment of mul-

³*Editors:* Alongside estimates of global and regional poverty totals based on the International Poverty Line of \$1.90 a day, the World Bank’s October 2017 release of the PovcalNet database provided figures for two additional poverty lines, \$3.20 a day and \$5.50 a day, where these lines are “based on the national poverty lines typically found in lower- and upper-middle income countries, respectively” (PovcalNet website, “What is new,” 10 October 2017). We remind the reader that the dollars concerned are international purchasing power dollars.

tiple poverty lines provides one test of the sensitivity of the estimates. There can be greater confidence in the monitoring exercise if a range of poverty lines around the International Poverty Line all show a decline in the percentage of the population considered poor: there is then “dominance” (Atkinson 1987). However, simply taking a mechanical multiple does not seem to be the right course, in that there is no evident justification for taking a particular multiple. It would not satisfy those who are critical of the existing \$1.90 a day line. It would adjust all country lines by the same percentage, whereas the relativities are open to question. As put to the Commission on Global Poverty by ATD Fourth World, “tracking poverty on other lines, such as \$4 or \$10, would just replicate the same problems and weaknesses” (2015, p. 5). One (although not the only) input into the choice of alternative lines should be information obtained by participatory studies. Do those living around the extreme poverty level consider that it is realistic? What is the view of those living in countries where there are many people in extreme poverty?

The second application is to learn more generally from participatory studies about the design of nonmonetary indicators of poverty. One aim of these indicators is to give voice to those individual circumstances that are not brought out by household-based measures, particularly the position of women and children. The Global Coalition to End Child Poverty has stressed that “participatory methods allow us to capture the views and perspectives of children, young people and other vulnerable groups providing valuable insight to better understand and contextualize poverty, vulnerability and exclusion” (2015, para. 1.1). Referring to the need for greater efforts “to ensure that all children are counted as part of poverty assessments, including children living outside of households and/or without parental care,” the Coalition goes on to say that “nothing reveals this more clearly than listening to children themselves, about what they need and what they want.” Or as Richard Jolly puts it, “Child poverty is . . . very different from adult poverty defined in terms of living below \$1.25 a day [then the International Poverty Line]. . . . For a child, . . . adequacy must be assessed not just in terms of a child’s immediate needs but in terms of what the child needs to grow in strength and capabilities so as to reach adulthood, with the basic capabilities needed for being a good citizen, within the community and beyond” (2012, p. xxxi). I discuss ways of taking more account of the position of women and children at greater length in chapter 3.

Conclusions

There is much that can be learned from participatory engagement, and poverty standards should be set after conversations all round the world, not just in Washington. However, subjective assessments cannot be the sole basis for monitoring trends in poverty. Otherwise, we would have to accept that a country can become poorer even though nothing has changed as far as an external

observer is concerned. We need in addition measures of poverty based on objectively observed characteristics of households, to which I now turn.

CONSUMPTION AND BASIC NEEDS

What do we observe about households? The World Bank global poverty estimates are based on the level of household consumption as recorded in household surveys. (The concept of consumption is dealt with in chapter 3, and how it gets recorded is discussed in chapter 4.) This observed consumption is then compared with that deemed necessary to avoid poverty, but setting that poverty standard has been one of the central challenges to research in this field.

I start where much of the scientific study of poverty began: relating the poverty standard to a concept of *basic needs*. Such a concept has the apparent merit of providing a physiological foundation for the poverty measure. As I shall argue, this appeal to physiology is overdrawn, but the approach has long attracted investigators of poverty. At the beginning of the twentieth century in the United Kingdom, the chocolate manufacturer Seeböhm Rowntree (1901) drew on the work of early nutritionists to determine minimum food requirements for a household, to which were added other “necessaries,” consisting of clothing, fuel, and sundries, and an addition covering the rent paid. In the United States, an early application of linear programming was to the derivation of minimum cost diets that met specified nutritional requirements (Dorfman, Samuelson, and Solow 1958, p. 9n).⁴ The official US poverty line developed from the work of Mollie Orshansky (1965), who took as her starting point the estimates of minimum food expenditure by the US Department of Agriculture, examined the proportion of income spent on food in households of different types and then multiplied up the food spending to allow for nonfood items. In India in 1962, the Planning Commission set a minimum consumption level as a target for the fifth Five Year Plan, based on the “minimal level” diet published by Sukhatme (1961), multiplied up to allow for nonfood spending.

Today a basic needs approach is widely employed in the derivation of national poverty lines, as shown by many examples in the national reports at the end of this book. In Jamaica, for example, the poverty line has been defined in terms of a food basket designed to provide a minimum nutritional requirement for a family of five, with an addition being made for nonfood items to cover the cost of clothing, footwear, transport, health and educational services, and

⁴Linear programming is a mathematical technique to obtain the minimum value of a linear function (in this case the cost of the diet) subject to linear inequality constraints (providing at least a specified level of each nutritional requirement). Stigler (1945) had previously solved the problem by a trial-and-error process that came close to the full linear programming solution. The application of linear programming in the definition of poverty lines for global poverty estimates has been revisited recently by Robert Allen (2017) in work discussed further in chapter 3.

other personal expenses (Planning Institute of Jamaica 2007). In Italy, the national statistical institute regularly produces poverty estimates based on baskets comprising food, clothing, housing, and other necessities which vary by age and number of household members and which are valued accounting for differences in prices across geographical districts and between urban and rural areas (Istat 2009).

The basic needs approach

What does the basic needs approach involve? There are three key steps in the method as typically practised:

1. Determination of nutritional requirements, here discussed in terms of individual food calorie requirements per day, which may vary with age and gender
2. Conversion of these requirements into a food budget, which involves the nutrition content of individual foods and their cost
3. Making an allowance for nonfood items

The first step draws on a large body of research by, among others, the World Health Organization (WHO) and the Food and Agriculture Organization (FAO) on calorie requirements. Calories are probably familiar to readers from information on the backs of cereal packets, but it may be helpful to spell out that they are a measure of energy requirements. I shall be referring to calories with an upper-case C (Cal.), which is the energy needed to increase 1 kilogram of water by 1°C at a pressure of 1 atmosphere. (You may also see references to calories with a lower-case c, which replaces 1 kilogram by 1 gram; thus, 1 Cal. = 1,000 cal.)

At the same time, the first step is not just a matter of looking up a number in a table. The Calorie levels required depend on the level of physical activity assumed. The FAO (2001, p. 36) provides estimates of energy requirements. These are defined relative to basal metabolism (“sleep”) at 1.0 and range from 1.53 when the main daily activity is “sedentary or light” through 1.76 (“active or moderately active”) to 2.15 (“vigorous or vigorously active lifestyle”). In more detail, “sitting on a bus or train” is scored at 1.2 for both males and females, whereas pulling a two-person rickshaw is scored for males at six times greater, 7.2 (FAO 2001, p. 92). Thus there is a range of choice for energy needs, and in practice different choices are made when constructing poverty standards. If we look at the Calorie requirements underlying national poverty lines in different countries, then we find that in Sri Lanka they are 2,030, in South Africa 2,261, in Pakistan 2,550, and in Uganda 3,000 (World Bank 2017, figure 2.1).⁵

⁵ *Editors:* The figure for South Africa refers to 2001. The national report for South Africa at the end of the book cites a figure of 2,100 Calories being used in the official poverty line, which dates from 2012.

Such a range underscores the fact that judgement enters at each stage of the process. That the judgement may be questioned is illustrated by the World Bank's comment on the official poverty line in the former Soviet republic of Georgia, a line based, the Bank argued in 1999, "on the (unrealistically) high nutrition norms inherited from the pre-independence period" (World Bank 1999, p. 1).

The second step is the conversion of the nutritional requirements into food purchases, where the purchases are determined to secure the specified requirements of Calories and other nutrients at the lowest total cost given the price of each foodstuff. It follows that the composition of the food basket, and the total outlay, depend on the prevailing food prices and availability. If supermarkets in one country raise the price of flour, then this may tilt the basket away from flour, and it will certainly raise the total cost. It is also evident that the application of the second step necessitates the collection of detailed data on the cost of different foodstuffs.

This may sound like a mechanical process. However, the diet produced by the early linear programming calculations in the United States led to questions as to whether it was realistic to assume that it would be followed in practise. The application of mathematics took no account of whether people would actually eat the proposed diet. In his pioneering UK study, Rowntree (1901) drew on actual dietaries used to feed people in institutions, and so he included tea in the list of goods, even though it has no nutritional value. The minimum food expenditures used by Orshansky (1965) were based not solely on the intake of Calories and other nutrients, but also on views regarding an acceptable trade-off between nutritional standards and consumption patterns. These departures mean that the poverty line is set higher, to a degree that depends on judgement.

For this reason, an alternative approach may be followed, based on actual food consumption and the extent to which it generates the required nutritional intake. In simplified form, this means examining the nutrients contained in the foods consumed by different households (which requires information on the quantities and qualities of foods, such as in the case of Sri Lanka grams of basmati rice, as well as the amount spent) to see at what level of total food consumption the required intake is attained. Reading across from nutritional intake to total food consumption gives the food poverty line. The national reports at the back of the book show that many countries use the resulting figure as a threshold for what is often labelled as "extreme" poverty—the situation where total household expenditure on both food and other items is below the food poverty line, as in, for example, Ghana, Kenya, Zambia, Bolivia, Brazil, Colombia, and the Dominican Republic. In Kenya, the National Bureau of Statistics refers to this as "hard-core" poverty and in addition defines "food poverty" as a household's food expenditure alone coming below the food poverty line.

The third step involves the addition of nonfood items; here there is a choice of methods and considerable room for discretion. Following the approach in the previous paragraph, we could read across, not to total food consumption, but to total overall consumption, including nonfood. A total poverty line would be set by the amount of total consumption at which a household could be expected to spend just enough on food to satisfy the nutritional requirements. For the household just at the nutritional cutoff, this would be equivalent to multiplying the food consumption by the reciprocal of the food share—i.e., if the household spends half its consumption on food, then we would multiply by 2. Alternatively, as in the work of Orshansky in the United States, the food share may be estimated from another source; the value selected was 0.33, so the cost of the minimum food expenditure was multiplied by 3. The food share is taken as 0.5 in Brazil and as 0.7 in Georgia, leading to the cost of the food bundle being multiplied up by 2 and 1.43, respectively. Or nonfood needs may be estimated directly, specifying minimum requirements of clothing, shoes, heating, health care, and other items, as in the most recent poverty line used by the Planning Commission of the Government of India, although in this case the difficulty of setting norms led the Expert Group to recommend that in practice observed expenditure on these items by households in the middle of the distribution be taken instead. In the EU, there has been extensive research on consensual budget standards. (See, for example, Bradshaw and Mayhew 2011, which contains a synthesis report covering thirty-two European countries.) A major project has been the EU Reference Budgets Network (see Goedemé et al. 2015; Storms et al. 2013). While a “consensual” standard provides a line of justification, it is rather removed from the concept of the poverty line having a purely physiological basis. As Peter Townsend remarked in his critique of the basic needs approach in the United Kingdom, “A family might maintain its physical efficiency just as well in a caravan . . . as in a three-bedroom house. It could go to bed early and spend nothing on electricity” (1962, p. 215).

So far, I have discussed the calorie requirements as though everyone were identical, but the construction of the basic needs standard has to make judgments regarding its *structure*. An important example is provided by gender. Should the basic needs standard be the same for men and women? One consequence of the nutritional approach is that the requirements are usually set at a lower level for women than for men. Applying the basic needs approach as described above would mean setting a lower poverty line for women than for men (as used to be the case in the United States and is still the case in Jamaica but not in Italy, to use the examples given at the start of this section). Is this a feature that should be introduced into the global poverty calculations? Given concerns about the gender composition of the poor, such a move would spark definite reactions, and the application of the basic needs approach may have to be accompanied by an overriding condition that the resulting poverty standard be gender-neutral.

To put some flesh on this account, let me illustrate by the calculation made for Vietnam in constructing the basic needs–based poverty line for 1993. (The procedure has since been revised, but the earlier version conveys the essence more clearly.) As described, in setting the Calorie requirement at 2,100 per day based on WHO and other sources, “it is useful to take an average requirement over the entire population” (World Bank 1999a, p. 145): i.e., to take no gender differences into account. To identify a basket of food consumption that meets this calorie target and reflects Vietnamese food consumption patterns, households in the 1993 Vietnam Living Standards Survey were divided into five equal-sized “quintile” groups in order of their total expenditure per head.⁶ For example, those households ranked from 40 per cent to 59 per cent made up the middle quintile group. It turned out that it was this group whose food consumption generated calories per capita closest to the 2,100 Calorie threshold, and it was their consumption that was used to identify the basic-needs basket of foodstuffs. So, if one looks into the entrails, one learns that the basket contains, among other things, 169.6 kilograms of ordinary rice per year, 11.4 kilograms of sweet potatoes, 5.9 kilograms of cabbage, and 6.6 kilograms of bananas (World Bank 1999a, p. 148, table A.2.2).

The household survey collected data by commune on the market prices of nearly all the relevant foodstuffs, and the median prices were used to make a calculation for Vietnam as a whole. The resulting total for January 1993 was 749,723 Dong, and this was the basis for the Vietnamese Food Poverty Line. To apply it to the measurement of poverty, the recorded household expenditure had to be adjusted to a national basis, adjusting for regional price differences, and to a January 1993 basis. To go from the Food Poverty Line to the General Poverty Line, a simple addition was made for nonfood based on the average nonfood expenditure of the third quintile group. This brought the total to 1,160,363 Dong per person per year in 1993—i.e., an increase of 55 per cent over the Food Poverty Line, reflecting the high share of food in total spending in Vietnam at that time (around two-thirds). It should be noted that the nonfood expenditure figure included both explicit spending and the imputed values of durable goods and owner-occupied housing. (I return to this issue in chapter 3.)

Basic needs as a moving target

The above description relates to the establishment of the basic needs–based poverty line at a particular point in time. But how should it change over time?

⁶A distinction should be made between quintiles and quintile groups. “Quintile” refers to the level marking the points in the distribution when we reach 20 per cent, 40 per cent, 60 per cent, and 80 per cent. The “quintile groups” consist of those between 0 and 19 per cent (first quintile group), between 20 and 39 per cent (second quintile group), and so on.

The simplest answer is that the line is increased in line with the general level of prices, as measured by the consumer price index (CPI). “Simple” is, however, a misnomer, since such price adjustments are often much debated. Does the CPI, designed for the general population, capture the effects of inflation on the poor? Is the CPI itself fit for purpose? Price indices have often been the subject of intense political disagreement. These questions are taken up later in chapter 4.

Even, however, if there is agreement on the price adjustment, experience with the basic needs approach has shown that in application it is subject to continual revision. The approach described above in the case of Vietnam in 1993 has since been revised, with a broader consumption aggregate and a different technique for obtaining the required total food spending. In Malaysia, the official poverty line was first formulated in terms of food needs in 1977 and was used until 2005 to monitor the country’s progress in eradicating poverty, but in 2005 the line was substantially revised to make it more comprehensive and more generous than its 1977 predecessor. In Nepal, the poverty threshold was defined to satisfy minimum caloric requirements, but in 2010 it was replaced by one that was 35 per cent higher in real terms “to reflect changes in well being over time” (Nepal Central Bureau of Statistics 2011, p. 16). In Italy, the method used to estimate absolute poverty since 1997 was extensively revised in 2009 by an ad hoc commission which recommended reviewing periodically the consumption basket and the hypotheses adopted in its definition in order to reflect changes in “the socio-economic reference context” (Istat 2009, p. 24). Going back in the mists of time to the studies of poverty in the United Kingdom by Rowntree (1901), his later poverty lines applied in 1936 and 1950 allowed for additional nonfood items, such as a radio, that had not been included—or indeed contemplated—in his first study of 1899.

The official US poverty line provides an exception to the policy of revision. The poverty thresholds for different types of family are adjusted each year by changes in prices, but the thresholds themselves have been left unaltered (bar revisions in the case of farm households, for example). Poverty in the United States today is still assessed by the Census Bureau on the basis of a calculation made in the first half of the 1960s. The use of this line in the administration of benefits has contributed to curbing its updating, despite the steady erosion of its value relative to mean income (Blank 2008). The Census Bureau also presents figures that incorporate regular changes in the consumption basket taken to represent basic needs, but, significantly, they are labelled a “supplemental poverty measure.” (The development of the US poverty line and its history in the following twenty-five years is described in Fisher 1992; the supplemental measure is explained in Renwick and Fox 2016 and followed on from the review of poverty measurement in the United States by Citro and Michael 1995.)

Adjustments to the basic needs standard may be expected to be upwards in the case of nonfood items, as in the examples just cited, but on what are they

based? How do we decide on an extension of the list of items? One route is via subjective assessments, although, as noted earlier when discussing the level, this takes us away from a physiological basis; moreover, the specification of the poverty line is inevitably influenced by prevailing living standards and the possibilities of the technology of the time. Can this be avoided by the use of food requirements, multiplied up by the inverse of the food share? Not entirely, since the question then becomes, whose food share? One of the few empirical regularities in economics with a lengthy shelf life is Engel's Law, which states that the proportion of total spending going to food declines as total spending rises. (This was the work of the German statistician Ernst Engel, not to be confused with Friedrich Engels.) As a society gets richer and the food share falls, the reciprocal—the inverse of the food share—rises, and with it the implied allowance for nonfood items associated with any set of basic food needs.

The food needs themselves raise major questions when it comes to updating over time. The fact that energy requirements are greater for those undertaking heavy work means that the food requirements should vary with the level of activity. Put in reverse, a country where most workers are sitting looking at screens should have a lower nutrition requirement than one in which many people are pulling rickshaws. Allen (2013, table 2) shows a time budget for a labourer in mid-eighteenth-century London, whose activities required 2.16 times the intake of the basic (sleeping) requirement. If the worker had instead been seated at a desk, the requirement would have fallen to 1.41. The lower needs may be offset in part by the fact that nutritional requirements rise with the greater height and body mass in rich countries, but the possibility should be contemplated that a basic needs-based poverty line should fall the higher the level of development. Such an outcome would no doubt puzzle those unfamiliar with the construction of the poverty line and might cause political embarrassment. It would be possible to specify that adjustments should be “upward only,” but that would be a flagrant departure from any physiological justification.

That this is a potentially serious matter is illustrated by the case of India. The pattern of calorie intake in India has been examined in depth by Deaton and Drèze, who find that “average calorie consumption in rural areas was about 10% lower in 2004–05 than in 1983. The proportionate decline was larger among better-off sections of the population, and close to zero for the bottom quartile of the per capita expenditure scale” (2009, p. 62). On the face of it, this evidence points to a lower level of basic needs associated with less demanding activity levels, although they conclude that “this hypothesis remains somewhat speculative.” Even more importantly, they argue that “average calorie intake has serious limitations as a nutrition indicator [and] close attention needs to be paid to other aspects of food deprivation, such as the intake of vitamins and minerals, fat consumption, the diversity of diet, and breastfeeding practices” (2009, p. 62). I come back to this in chapter 3 when discussing the design of

multidimensional measures and the use of outcome indicators such as stunting.

Inverting the telescope

Before leaving the basic needs approach, I should point out that we can “invert the telescope.” The discussion so far has run from basic needs to the resulting poverty line, but the approach is also illuminating in the reverse direction. The question becomes, what does the International Poverty Line of \$1.90 per person per day allow a household to buy? By relating the poverty standard to household budgets, light can be cast on its implications for household living standards. There was a famous occasion in the United Kingdom during an inquiry into dockworkers’ pay in the 1920s, when the union leader appeared in court with a plate bearing a few scraps of bacon, fish, and bread and asked the statistician (Sir Arthur Bowley) whether this was sufficient breakfast.⁷

Today the basic needs calculations would allow different levels of total consumption to be mapped back to the level of nutritional intake that they permit. The calculations would allow scrutiny of the allowance for nonfood items. In the case of India, Drèze and Sen note that the Indian “reference budget associated with the urban poverty line includes . . . ten rupees per month for ‘footwear’ and forty rupees per month for health care. The former would just about make it possible to get a sandal strap repaired once a month, and the latter might buy something like the equivalent of an aspirin a day” (2013, p. 189). Subramanian has similarly commented that in urban Tamil Nadu the official poverty line for a family of four, after allowing for shelter and food, “would have left a good deal less than nothing to spend on education, clothing, and transport” (2009, p. 68). The reverse calculations provide a basis for engaging the wider public in a consultative process: “Such an approach would make more meaningful the otherwise arcane statistical procedures on which the risk-of-poverty indicator is based. It would be a good means by which Governments could engage those experiencing poverty and social exclusion” (Marlier et al. 2007, p. 156).

Conclusions

The basic needs approach has played a major role in the development of poverty lines around the world. Historically, it has been influential in the measurement of poverty in rich countries. It provides insight into the meaning of

⁷*Editors:* The work of artists Stefen Chow and Lin Huiyi provides contemporary examples of poverty-line diets. They have taken photos for some thirty countries of food that can be purchased on the budget provided by national poverty lines, photos drawn on for the cover of this book (see www.thepovertyline.net).

poverty lines derived by other means—illuminating what the International Poverty Line, for example, would buy. However, the claims for the basic needs approach should be stated carefully. It cannot be said to provide, even at a point in time, a purely physiological foundation for measuring poverty, since at each stage of the calculation a significant degree of judgement is being exercised. It can be implemented in a variety of ways. When it comes to up-rating over time, both the nonfood and the food components raise major issues, which is a serious obstacle to its use for monitoring progress.

POVERTY AS DEPRIVATION OF CAPABILITIES

If the basic needs approach is long-standing, the capability approach dates from recent decades. In a series of contributions, Amartya Sen (1985, 1992, 2009) has argued that well-being should be judged in terms of the *functionings* achieved by people and of the *capabilities* open to them. By functionings are meant the activities and states valued by a person, and capabilities are the various combinations of functionings that he or she can achieve: “the actual opportunities of living” (Sen 2009, p. 233). In the context of measuring poverty, Sen has argued that this should be seen as the deprivation of capabilities, where that deprivation limits the freedom of a person to pursue their goals in life: “Identifying a minimal combination of basic capabilities can be a good way of setting up the problem of diagnosing and measuring poverty” (1993, p. 41). In this section, I consider what this means and how it can be put into operation.

The capability approach

Reactions to the capability approach vary a great deal. On one side there is enthusiastic endorsement among those who see it as offering a new perspective to assessing well-being and as transforming the measurement of deprivation. On the other side, there are those who question whether the vocabulary of capabilities adds much that is new; they suggest that the implications for poverty measurement are of little significance. It is with the latter that I begin. What is meant by the capability approach, and how does it differ from that based on household consumption, as in the textbook account?

There are in fact several key ingredients. The first is a concern with *opportunities*, where these describe the scope for choice but are not confined to the actual choices made. The capability set is a description of the menu of possibilities. A young person may dream of playing for Barcelona or at Wimbledon when they grow up, but more realistically they may plan to become a teacher, or to set up their own business as a computer consultant, or to become an independent fisherman. If they know that their family cannot afford to send them

to school, or that they have not the capital to buy a boat, then this is a restriction of the range of opportunities. Now, it may turn out that the person wants to be a teacher in any case, so the absence of capital to buy the boat is irrelevant. The achieved functioning is the sole concern, and we are back with an evaluation close to that applied in the standard approach. However, the essence of the capability approach is that we are in addition concerned with the roads not followed, and particularly with those that are not followed because they are blocked. As it was put clearly by Kenneth Arrow with the aid of some mild algebra, “The choices that are open to the person but that she or he does not make are also relevant. . . . The objects that should be ordered are not the outcomes x but the pairs (x, A) , where A is the set from which the choice is to be made [‘menu’ in Sen’s terminology] and x belongs to A ” (2006, p. 54).

Evaluating the outcome is the meat and drink of what this book is about, but what about the set of opportunities? The chances, in the example, of entering different occupations are multiple valued, not immediately reducible to a single number. How do we combine the chance of being a teacher and that of being a fisherman? In a sense, economists are used to doing this. If we have multiple indicators of deprivation, then the same problem arises. One resolution is to look for comparisons that are unambiguous, where, for example, a country is doing better, or no worse, on all dimensions. In the case of a person, a new opportunity, such as the opening of a college for nurses, unambiguously expands the set of options. The new situation is said to “dominate” the previous one. But what it does not resolve is what happens when the opportunities for nursing improve but pollution means that fishing is a less viable option. We have then to combine the different elements via some procedure for aggregation to see whether the gain from one outweighs the loss from the other. This procedure will take into account the relative desirability of the different career paths, and this begins to sound like the kind of process lying behind the determination of the actual outcome. Is it not much the same as a household choosing its consumption to maximize its well-being? In fact, the process is different, since the implementation suggested by Arrow and others takes account of the value of flexibility where the valuation is made in advance of knowing the actual preferences and the valuation is not solely based on the actual outcome. There are differences, but they are of some subtlety. The capabilities theory does indeed illustrate the general point that the goal of securing equality of opportunity, while enjoying wide acceptance, is a concept whose precise definition is varied and often elusive, whereas the goal of tackling outcome deprivation, much more contested, is—relatively—easier to translate into agreed concrete measures.

It is to concrete measures that I turn below. First, however, there is a second major ingredient, one that links with other developments in consumer economics, notably those that see the household as akin to a small factory, engaged in household production. The household budget approach to poverty measurement

has pinpointed the consumption of goods and services (“commodities”), but Sen, like a number of other economists, has argued that commodities are a means to an end and that we need to consider the ways in which they make possible the functionings which are the ultimate concern. He traces therefore the chain

Commodities → Characteristics → Capabilities → Well-being

So, to take Sen’s example, possession of a bicycle has the characteristic of allowing you to travel; this characteristic ensures the capability to get to work, and that underpins your well-being. Where the capability approach differs is that it is concerned with the third, not the fourth, link in the chain. In contrast, the basic needs approach starts from a specified standard of living and works back to the necessary commodities, with little attention being paid to the intervening steps.

A focus on the intervening steps is a crucial contribution of the capability approach. To see what is involved, let us suppose that each unit of commodity—which may be food or nonfood—yields a specified amount of each of the relevant characteristics. The amounts per unit of a commodity are referred to as “output coefficients,” and the whole array covering all commodities makes up a “matrix.” In the case of foodstuffs, the coefficients tell us how many calories, proteins, vitamins, etc., are generated per unit purchased. A set of purchases will then provide a total, adding up across all the commodities, of each of the characteristics. Since the characteristics are what secures the desired level of capabilities, we want to reverse the process and find the necessary purchases of commodities. In mathematical terms, this requires us to invert the matrix, so we can read back from the desired level of capabilities.

In terms of foodstuffs, what has been described so far is not different from the linear programming exercise used in the early US calculations, but there are two distinct features that become apparent from the capability approach. The first is that the coefficients are not purely technical. The significance of such an approach is demonstrated by Sen in his analysis of “Gender and Cooperative Conflicts” (1990). He builds on the household production approach, but with the key difference that “technology is not only about equipment and its operational characteristics but also about the social arrangements that permit the technology to be used and the so-called productive processes to be carried on” (1990, p. 463). In terms of foodstuffs, the social arrangements clearly include such matters as dietary norms and religious restrictions, but for all commodities it is the case that the coefficients reflect the prevailing consumption practices in the society in which people are living. In his 1899 investigation, Rowntree recognised that the allowance for clothing for a worker should “not be so shabby as to injure his chances of obtaining respectable employment” (1902, p. 108). The allowance today would be rather different. The matrix of coefficients depends on the society and the date.

An immediate consequence is that the commodities required to achieve a specified level of functioning are influenced by the society in which a household lives and may change as the society develops. As it was put by Sen, “In a country that is generally rich, more income may be needed to buy enough commodities to achieve the *same social functioning* . . . to the capability of ‘taking part in the life of the community’” (1992a, p. 115). If the level of consumption that is required to function within a society varies with the average level of consumption, then a standard that is *constant* in the space of functionings or capabilities would imply a standard in terms of consumption that varies over time and across countries.

There is a second dimension of poverty that has so far been missing from the story: the role played by suppliers and producers. In constructing the matrix of coefficients, account has to be taken of the extent to which goods and services are *available*. Households can only buy what is in the market, and this is decided by the suppliers. Potential consumers may be excluded from the market by the decisions of firms regarding which quality of product to supply. As overall living standards rise, shops may no longer offer cheaper cuts of meat, and small-sized packs of food may no longer be available. The cost of food may depend on the location of employment: a worker may have to purchase more expensive meals away from home on account of the increased concentration of production in distant urban locations. A worker dependent on public transport to get to this work may find the service withdrawn as the rest of society gets richer and can afford its own transport.

Capabilities and poverty measurement

The capability approach is in many respects congruent with other developments in the measurement of poverty in rich countries that have stressed its role in preventing people from participating fully in society. This is indeed the way in which poverty has been defined following the work of UK sociologist Townsend, who saw poverty in terms of ability to participate in society: “Individuals, families and groups in the population can be said to be in poverty when they lack the resources to obtain the types of diet, participate in the activities and have the living conditions and amenities which are customary, or at least widely encouraged or approved, in the societies to which they belong” (1979, p. 31). Such an approach was implemented in 1984 by the European Council of Ministers, who defined those in poverty as “persons whose resources (material, cultural and social) are so limited as to exclude them from the minimum acceptable way of life in the Member State in which they live” (decision of the Council of Ministers, 19 December 1984). The capability approach provides theoretical underpinnings for this essentially political definition of poverty, but there is clearly a gap between the idea that the necessary commodities should rise with living standards and the concrete implementation in terms of poverty

measures. This gap is where, just as with the basic needs approach, the exercise of judgement plays an important role. The current form of the EU commonly agreed measure of “risk of poverty” draws a threshold at 60 per cent of the median disposable income (adjusted for household size and composition) in the country in question. But why 60 per cent? The 60 per cent of median threshold replaced an earlier line drawn at 50 per cent of the mean, but that simply pushes the search further back in time. The study underlying the design of the EU Social Indicators (Atkinson et al. 2002) referred to a range of percentages from 40 per cent to 70 per cent.

In seeking an answer to this question for the United Kingdom at the time (1968–1969), Townsend took an approach that at first seemed promising, which was to examine the relationship between observed income and the reported inability of a household to participate in a range of activities. The activities included such matters as having a week’s holiday away from home, having a birthday party for the children, having sole use of a flush WC, sink, bath, or shower or gas/electric cooker, or having a cooked breakfast most mornings (remember it was the 1960s). The resulting deprivation score, formed by combining these indicators, when averaged over households in an income range showed a fall as incomes rose. What Townsend was seeking was a threshold, or a kink in the relationship, with the decline being steeper below the threshold. The findings turned out to be suggestive but “inconclusive” (1979, p. 255). While there was “no evidence . . . of a pronounced threshold . . . the increase in the proportion of households with high scores was particularly marked once income had fallen to a level of 60 to 69 per cent of the mean. This suggested that if a poverty line were to be drawn, it would seem justified to draw the line within this band” (p. 258).

I have spent some time on this classic study because it is a fine example of applying social science research to answer a very real policy question. Fast-forwarding some thirty years, the EU-based study by Fusco, Guio, and Marlier (2010, figure 6.2) examined in a similar way the relation between the intensity of deprivation and household income in twenty-five European countries. The deprivation questions were not the same: ownership of a washing machine, colour TV, and a car were now entered. Some questions, on the other hand, were similar, such as a week’s holiday away from home. Inspection of the graphs (by eye) suggests that there was a smooth downward curve and no threshold in a number of cases: Greece, Hungary, Latvia, Lithuania, Poland, Portugal, and Spain. But for others, a threshold is detectable (subject to more rigorous testing) at around 60 per cent of the median income (adjusted for household size and composition): Austria, Belgium, Cyprus, Germany, Estonia, Italy, Luxembourg, Slovakia, and Slovenia. These findings are suggestive, but not definitive. How this evidence is weighed is a matter of judgement.

The capability approach may provide only an incomplete answer to the level at which the poverty line should be set, but it does, however, go further. In par-

ticular, it provides a framework for thinking about the *structure* of the poverty standard. This is demonstrated by Sen's lead example of disability: "The relevance of disability in the understanding of deprivation in the world is often underestimated, and this can be one of the most important arguments for paying attention to the capability perspective. People with physical or mental disability are not only among the most deprived human beings in the world, they are also, frequently enough, the most neglected" (2009, p. 258). Here lies the value of separating the links in the chain from commodities to capabilities. In the example given above, it was taken for granted that possession of a bicycle supplied the capability of getting about, but for people with certain disabilities, this is not the case. At the very least, the bicycle would have to be specially adapted. The technological coefficients are different, and hence different amounts of goods and services are needed to ensure a specified level of functioning. (I return to this issue in the next chapter.)

Finally, capabilities are essentially multidimensional, and the construction of multidimensional indicators may be viewed as an operationalization of the capability approach, where the theoretical foundation provides guidance as to the dimensions to be included. At the same time, just making a list of dimensions is not sufficient, and—as we have seen in the case of income poverty—the setting of concrete thresholds requires other inputs and the exercise of judgement.

Conclusions

The capabilities model has undoubtedly enriched the way in which we approach the measurement of poverty and provides a framework for tackling difficult definitional issues—including the roles of absolute and relative poverty, to which I come in the final part of this chapter. At the same time, it does not eliminate the need for judgement: there is a gap between theory and application. It is an approach, not a resolution.

MINIMUM RIGHTS AND POVERTY

To this juncture, we have been concerned with a well-being approach to the measurement of poverty. I turn now to the alternative "rights-based" approach, an approach that has elements in common with those already discussed, but which may take us in some different directions. If you doubt this and are thinking of skipping, I should just note that the basic needs approach in the United States happily for many years set a poverty line that was lower for women than for men, whereas that would have been ruled out of court by a rights approach.

Human rights

I begin with a compact account of the rights approach. “Compact” may seem inconsistent with the vast literature on “rights,” and indeed a necessary immediate step is to limit the field severely. First, I do not discuss the instrumental role of the right to freedom from poverty in underpinning a general catalogue of human rights. The abolition of poverty may be an essential precondition for the full enjoyment of other human rights, but it is with the intrinsic importance of attaining “poverty rights” that I am concerned here. Second, I return to a governmental view of the definition of poverty, as in the first section of this chapter, taking as given the way that rights are defined by national—and, more especially, international—bodies. Again, performance is being judged according to stated goals.

How can this be made concrete? Can the objective be formulated in terms that capture the essence? “Human rights always come to us in lists,” according to Nickel (2014, p. 219), and he goes on to say that “international law uses a standard list that mainly comes to us from the Universal Declaration of Human Rights. This list consists of seven families of human rights, the first six of which are found in the Universal Declaration.” (The Universal Declaration was adopted by the UN General Assembly in December 1948.) The sixth on the list is “social rights that require that people be provided with education and protected against starvation and severe poverty.” The Universal Declaration itself has a more extensive list. Article 25 states that “everyone has the right to a standard of living adequate for the health and well-being of himself and of his family, including food, clothing, housing and medical care and necessary social services,” and Article 26 adds that “everyone has the right to education,” with a specific reference to elementary education.

This formulation in the Universal Declaration does not lead us directly to a poverty line, but it is informative in three respects. The first is that rights are defined on an individual basis. It is true that Article 16 says that “the family is the natural and fundamental group unit of society and is entitled to protection by society and the State,” but there are no attached rights of families or households. In this book, I take rights as individual. It is, for example, concern with the individual rights of women that underpins the case I make for nonmonetary measures that allow gender inequality to be identified. It should, however, be recognised that, right from the start, the Universal Declaration was criticised as being too dominated by Western values with their individual focus. Eleanor Roosevelt, widow of American president Franklin D. Roosevelt and chair of the Declaration’s drafting committee in 1947, recalled in her memoirs that some had argued for a more pluralist approach, with one Asian member suggesting that the Secretariat might well study the fundamentals of Confucianism (UN 2017). A starting point of society as a whole, or of distinct population subgroups (defined, for instance, by geography or ethnic composition

or religion), may lead to a focus on the right to live harmoniously in such a society. As noted in the next chapter, this would lead to the extension of a portfolio of nonmonetary indicators of poverty to include dimensions of a different kind.

The second respect is that the list of rights in the Declaration makes frequent use of the word “everyone” and is explicit as to when rights are not neutral between individuals. Otherwise, the rights are universal. To reiterate a point made earlier, this means that there is no justification for the situation when, as was the case in the past, the US official poverty line, derived from a basic needs approach, was lower for women than for men. At the same time, there are exceptions. Article 25 goes on to say that “motherhood and childhood are entitled to special care and assistance. All children, whether born in or out of wedlock, shall enjoy the same social protection.” Here particular attention should be paid to the rights of the child (Pemberton et al. 2007), as embodied in the UN Convention on the Rights of the Child, a document which, to quote the former UN high commissioner for human rights Mary Robinson (2002), “adds value because it provides a normative framework of obligations that has the legal power to render governments accountable.”

The third contribution is that the lists identify key dimensions that are relevant when considering a multidimensional approach, as in the next chapter: education, food, clothing, housing, medical care, and necessary social security. Such a list is similar to that given by rights theorists, although with some interesting differences. For example, Shue cites “unpolluted air, unpolluted water, adequate food, adequate clothing, adequate shelter, and minimal preventative health care” (1996, p. 23). Here again the focus on the rights of the child is important. “Child mainstreaming” has been stressed in the development of nonmonetary indicators in the EU. Eric Marlier, Bea Cantillon, Brian Nolan, and I commented that the common social indicators were unsatisfactory when viewed from the child perspective; instead, “we need to approach the issue from the opposite direction: to start from the perspective of children and then consider the selection of indicators” (2007, p. 102). (For further discussion of child material deprivation, see Guio, Gordon, and Marlier 2017; see also Micklewright 2002.)

Rights and poverty measurement

[*Editors:* Tony left no notes on this subsection. We guess he would have developed further the concrete implications of the Universal Declaration for poverty measurement beyond the three general issues just discussed. For the UN special rapporteur on extreme poverty and human rights, extreme poverty “involves a lack of income, a lack of access to basic services and social exclusion” (website of the Office of the UN High Commissioner for Human Rights, January 2018). See also UN (2012). Tony might have discussed the approach taken

in the reports of visits to countries in recent years by the UN special rapporteur. (These countries include Chile, China, Ghana, Mauritania, Romania, Saudi Arabia, the United Kingdom, and the United States.)]

*Rights and diverse cultures*⁸

The word “poverty” has different meanings in different cultures: “To write about the Japanese poor in English is an anomaly in itself, because whatever stands for ‘poor’ in the Japanese language has its unique flavour within the Japanese sociocultural context, lost when rendered into English” (Chūbachi and Taira 1976, p. 433).

A report by the Solomon Islands National Statistical Office and the UNDP puts it well:

Traditional Solomon Island society, as well as Pacific societies generally, embraces caring for and sharing with family and clan. As a result, there is a continuing belief that poverty cannot and should not be a part of normal life in the Pacific region. The suggestion that there might be poverty in some form is not, therefore, something that many people have been prepared to readily accept. Indeed, the usual images of poverty (starving children, landless peasants, and men and women toiling with ox ploughs) do not immediately spring to mind in relation to the Pacific or Solomon Islands. (2008, p. 8)

The report goes on to say: “However, the increasing monetization of Pacific economies, the impact of television and internet, and increasing rural/urban migration leading to greater urbanization, have begun to undermine these traditional structures. . . . As a consequence poverty and hardship, as now defined and understood in the Pacific, . . . are being increasingly accepted as concerns which need greater attention” (p. 8).

Returning to the capability approach, Martha Nussbaum has proposed functional freedoms, or central human capabilities, as a rubric of social justice, paying especial attention to the unequal opportunities of women in different cultural settings (for example, Nussbaum and Glover 1995; Nussbaum 2000, 2000a).

ABSOLUTE AND RELATIVE POVERTY IN A GLOBAL POVERTY COUNT

The different concepts described in this chapter have differing implications for the measurement of poverty. These will become apparent in the next chapter

⁸*Editors:* We have only isolated paragraphs for this subsection. Tony seems to have intended to discuss what poverty means in different cultures, as his opening words show.

when I discuss the translation of concepts into measures. But there is one important issue where we need immediately to confront the differences: the long-standing debate contrasting absolute and relative poverty. This acquires particular salience here where we are concerned with poverty across the globe, with countries at very different levels of development.

Absolute and relative poverty

To begin with, we must define terms. “Absolute” can bear at least two different meanings. The first is that there is an inherent justification for a particular poverty standard, with scientific basis in physiological or other considerations. It may require measurement, but in this respect is like the speed of light. As we have seen when discussing the basic needs approach, such a line is not readily established, and the idea of a purely physiological basis is illusory. The term “absolute” has come therefore to assume a second meaning: a poverty line that is fixed in terms of consumption, adjusted over time in line only with changes in prices. To distinguish this version, we may refer to it as “anchored absolute,” and we have seen several examples in this chapter. If we turn to a “relative” poverty line, then we find that the poverty status of a household is related to the prevailing living standards in the society in which the household lives. An example is where the poverty line is set as a percentage of the median consumption or income level of the country in question. There may be a mix of absolute and relative lines, where the initial poverty line is set as a percentage of the median, but the line is then adjusted over time according to the movement in prices, giving an “anchored relative” poverty line. This is the case with the lines used by the EU to estimate the “at-risk-of-poverty rate anchored at a point in time” (Eurostat 2017).⁹

Two features are apparent. First, an absolute measure, unlike a relative measure, can be applied to a single household in isolation. A person’s ability to meet his or her basic needs can be judged independently of the situation of all other inhabitants of the planet. In contrast, a relative measure, as with the capability approach, is essentially concerned with the ability to function in a particular society. Second, while there are extremes of pure absolutism and pure relativism, there are also intermediate cases along the continuum:

⁹*Editors:* Another way of achieving the mix of absolute and relative concepts is when an absolute line is indexed to movements in nominal incomes rather than prices, as, for example, with the social and subsistence minima calculated for the former Czechoslovakia in the 1970s (Atkinson and Micklewright 1992, p. 211). For a treatment by Tony of the distinction between absolute and relative lines in the context of measuring poverty in Europe, see Atkinson (1998, pp. 18–28). See also Foster (1998) for a discussion of different meanings of the word “absolute” in the context of poverty measurement.

Absolute line ↔ Anchored absolute line ↔
Anchored relative line ↔ Relative line

It should be evident from the earlier discussion that the different concepts point to different places in this continuum.

A global poverty measure

Following the global scope of the Sustainable Development Goals, the measurement of poverty is seen in this book as a global exercise, encompassing rich as well as poor countries. The embracing of both developed and developing countries, however, is often represented as generating an inevitable conflict between “absolute” and “relative” approaches. But this is not necessarily the case. The shift in perspective to a capability approach provides one route to resolve any such tension, as François Bourguignon and I have argued (Atkinson and Bourguignon 2000, 2001). Not only does the approach provide an underlying theoretical framework for analysing deprivation, but it also shifts the standpoint from which ethical considerations are applied.

The application of the capability approach can indeed be seen as building on the earlier discussion of the energy requirements of different activities. “Bare survival” may be seen as requiring for an individual a level of food and other inputs, broadly independent of the overall standard of living. On the other hand, “work” does indeed have higher energy requirements (with the caveat about the shift away from heavy labour), but it is an activity where other requirements apart from nutrition are significantly socially determined and can be expected to rise with the overall standard of living. A linen shirt may have been sufficient at the time of Adam Smith, but today a job seeker requires internet at home.¹⁰

This provides a way of understanding the evidence about national poverty lines. Many years ago, using data from 1985, Ravallion, Datt, and van de Walle (1991, figure 1) plotted national poverty lines against mean household consumption for thirty-three countries. Observing that the national poverty lines tended to rise with the level of mean consumption in the country in question, they fitted a statistical model to describe the relationship between the two, involving the logarithm of the poverty line rising with mean household consumption and its square. The same data, however, suggested to François Bourguignon and me (Atkinson and Bourguignon 2000, figure 1) that the national poverty lines were better viewed as—at that time—the greater of \$1 a day per

¹⁰*Editors:* The reference to the linen shirt is to Adam Smith’s famous comment in his 1776 *Inquiry into the Nature and Causes of the Wealth of Nations* that, “in the present times, through the greater part of Europe, a creditable day-labourer would be ashamed to appear in public without a linen shirt, the want of which would be supposed to denote that disgraceful degree of poverty” (book V, chapter II, article IV).

person in international purchasing power dollars and 37 per cent of mean consumption per person. There was a switching point around the overall consumption per head in Morocco. We went on to argue that such a switching explanation corresponded to the theory of capabilities described in the previous paragraph. Below a certain level of overall consumption, the availability of individual consumption associated with basic needs is what limits capabilities, but once the switching point is reached, it is another set of consumptions that limits capabilities. What is more, the first constraint is governed by a fixed level of consumption, appropriately adjusted for differences in prices, but the second set varies with the overall level of consumption in a particular context. Typically, this relativity is expressed as a percentage of the mean consumption in the country at a particular date.

This way of viewing the global poverty situation is well described in the report on the 2005 Kenya Integrated Household Budget Survey: there is a

hierarchy of basic needs which begins with survival food needs and is followed by basic non-food needs. Many activities that are deemed essential to escaping poverty cannot be performed without participation in society; for example, employment and schooling. That social participation is not possible without incurring the basic non-food expenditures on, for instance, shelter, clothing and hygiene. (Kenya National Bureau of Statistics 2007, p. 27)

The different possibilities are illustrated in figure 2.1, where countries' mean consumption per head is measured along the horizontal axis and an individual household's own consumption per head on the vertical axis. To determine whether a particular household is in poverty, it is necessary first to identify where their country is located along the horizontal axis—say, at the level X. The poverty status of the household is then governed by its own consumption per head: i.e., where the household is located on the vertical line going up from this point. For a country with mean consumption to the left of level A on the graph—the switching point—the applicable poverty line is the *absolute* threshold represented by the horizontal section of the poverty line, level B. In a country with mean consumption per head at the level of X, a household is in poverty if its consumption per head is below level B. For countries to the right of point A, the *relative* threshold, which is higher, is applicable, and this is represented by the upwards-sloping section of the poverty line. Among this group of countries we are concerned both with those households living below the absolute poverty threshold and with those above this threshold but below the relative poverty line. Thus, it is possible to identify three areas of poverty: areas I and II correspond to absolute poverty in poorer countries and richer countries, as identified by their positions relative to A; area III corresponds to relative poverty, a condition that occurs only in richer countries where escaping poverty requires satisfying needs that go beyond mere survival.

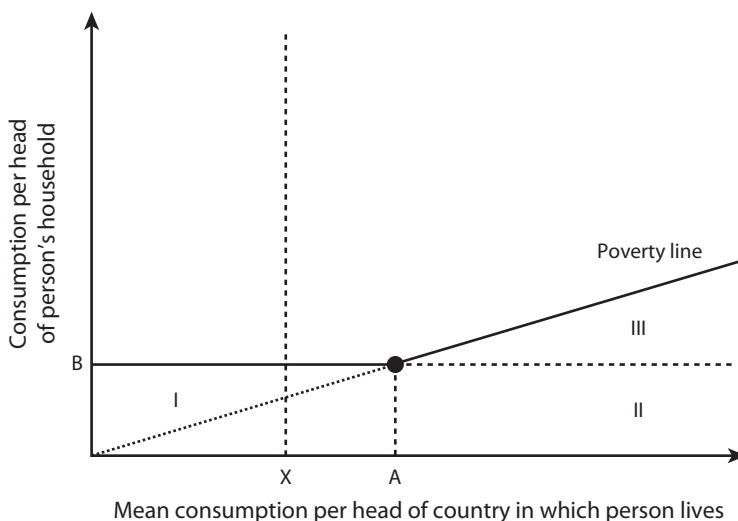


Figure 2.1. Absolute and relative poverty in global poverty measurement

Source: The graph draws on Atkinson and Bourguignon (2000).

Reading note: The poverty line has a fixed value, level B, until a country's mean consumption per head reaches level A, at which point the value of the line increases with the level of mean consumption per head.

This may sound complex, but the key constituents are familiar. The first is the absolute poverty threshold, level B, taken here for the purposes of discussion to be the International Poverty Line, \$1.90 per person per day. The second is the slope of the section of the line showing the relative poverty threshold. Here too there is a degree of familiarity, in that relative poverty measures in high-income countries are typically expressed as percentages of the mean or median income. So, in 2001, the EU adopted as its official definition of relative poverty—as one of the Commonly Agreed Social Indicators referred to as “poverty risk” (Atkinson et al. 2002)—60 per cent of the median disposable income (adjusted for household size and composition). This is broadly the counterpart of 50 per cent of mean income.¹¹ In other countries, the figure taken is 50 per cent of the median (Morelli, Smeeding, and Thompson 2015, p. 609), or closer to 40 per cent of the mean. The level of the absolute poverty threshold and the slope of the section of the line showing the relative threshold together determine which

¹¹ *Editors:* The distribution of individuals' household consumption or income is asymmetric with a long upper tail. The mean value (the arithmetic average) is some way above the median (the value which splits the distribution into two, with half of individuals above this level and half below).

countries are to the left or right of the switching point, A. The level of country mean consumption at which the regime switches is equal to the reciprocal of the slope times the value of the International Poverty Line. In the case illustrated, with a slope of 0.3, or 30 per cent, the poverty line starts moving upwards when a country reaches a mean consumption of three and one-third times the International Poverty Line. In 2011 PPP terms, this is a mean consumption level equal to \$6.33 a day. The figure of 30 per cent for the slope is close to the 37 per cent suggested by François Bourguignon and me (Atkinson and Bourguignon 2000). Question for the reader: how would the diagram have looked if the slope had been taken as 50 per cent?¹²

In chapter 9, I explore the implications of this application of the capability approach for the measurement of poverty in the world as a whole. In this way, we are making a reality of the goal of the SDGs to be inclusive of all societies—rich and poor.

Finally, it should be reiterated that the capability approach in the form adopted here misses much of the richness of the foundational work by Sen and Nussbaum, and that it involves a distinct shift in perspective. It is, for instance, possible that a rise in the income of a person below the poverty line leads to an increase in measured poverty. Seen in terms of income, this makes no sense (see Decerf 2015), but seen from the standpoint of capabilities, the consequent rise in mean income may have the consequence that people above but close to the poverty line are more challenged when they seek to participate in that society.

CONCLUSIONS

In this chapter, I have discussed different interpretations of the word “poverty,” staying at a broad, conceptual level. While there is a range of different approaches, I have tried to show that there is common ground that is relevant to any attempt to measure global poverty, with the capability approach playing an important unifying role in thinking about countries at different levels of development. The focus has been on the implications of each approach for the measurement of poverty, and the next two chapters now consider measurement in more detail.

¹²*Editors:* The answer comes in chapter 9! Using a much larger database in terms of countries and years, Jolliffe and Prydz (2017) reconsider Atkinson and Bourguignon’s approach together with alternative proposals by Ravallion and Chen (2011) and Garroway and de Laiglesia (2012). In the report of the Commission on Global Poverty, Tony discussed the possibility that the intercept implied from projecting back the sloping part of the poverty threshold is different from the value of zero shown in figure 2.1 (World Bank 2017, pp. 142–144), but here he went back to the original Atkinson-Bourguignon formulation.